Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	04 October 2026
Team ID	SWTID-2026-6870
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture

The solution is built on the ServiceNow platform using low-code tools. A Standard Laptop order triggers a Flow Designer flow which creates a catalog task for the Hardware team.

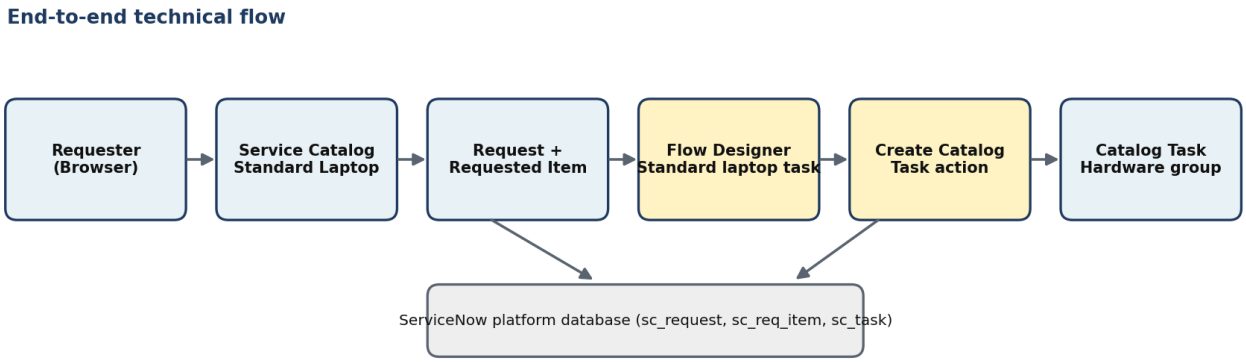


Figure 1: Technical architecture / data flow

Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How the user orders and tracks the laptop request	ServiceNow Service Catalog and platform forms (web browser)
2.	Application Logic-1	Automation that starts when the item is ordered	Flow Designer – flow "Standard laptop task" with Service Catalog trigger
3.	Application Logic-2	Creates the configuration task	Create Catalog Task action (ServiceNow Core spoke)
4.	Application Logic-3	Approval of the request before configuration	Request approval (Approvers related list), approver user
5.	Database	Stores request, requested item and task records	ServiceNow platform tables: sc_request, sc_req_item, sc_task
6.	Cloud Database	Database service on cloud	ServiceNow cloud-hosted instance database
7.	File Storage	File storage requirements	Not required for this project (platform attachments available if needed)
8.	External API	Calls to outside services	None used

S.No	Component	Description	Technology
9.	Infrastructure	Application deployment	ServiceNow cloud instance (Global application scope)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-source frameworks	The solution is configured, not coded; no external frameworks	ServiceNow Flow Designer (low-code)
2.	Security implementations	Role-based access, approval gate, flow runs as System User	ServiceNow roles, ACLs, Run As: System User
3.	Scalable architecture	Same flow design can be copied for other catalog items	Reusable flows and actions
4.	Availability	Cloud platform with managed availability	ServiceNow instance
5.	Performance	Automatic task creation, no manual step	Flow engine of the platform